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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Engineer with 3 years of industry experience owning production **GenAI** systems end to end at an asset manager: **RAG** over PostgreSQL with **pgvector**, tool-calling **agents** that run calculations and generate PDF reports, **FastAPI** microservices in Docker on **AWS** and a ReactJS frontend, shipped to Compliance, Client Service and Investment teams who relied on the output. Released to real users, iterated on what broke and had Compliance review each application before release; acted as de facto team lead for up to 4 interns and was twice recognised internally with Best Team Project (2024, 2025). Recently completed an MSc in Data Science at the University of Bristol, with a dissertation in partnership with AstraZeneca. Seeking a Senior AI Engineer role owning agents, retrieval and the systems around them for investment professionals.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Production GenAI: Agents, Retrieval & LLM Integration

- Designed and built **RAG** pipelines and knowledge retrieval for internal **GenAI** applications used across Compliance, Client Service and Investment (**LlamaIndex** with **Azure OpenAI GPT-3.5 & Llama-3**), using department-based retrieval: a separate **pgvector** database and **Azure OpenAI** embeddings for each department, so access stayed siloed and similar content from different departments was not confused.
- Built **agentic tool-calling** workflows where **AI agents** call **Python** functions as tools to run **SQL**-backed fund-comparison calculations and generate **PDF** comparison reports on user request.
- Built a **Python** extraction pipeline that turns over **300** unstructured daily emails from credit rating agencies into structured data, fuzzy-matches them to tracked companies and sends analysts a daily summary, **saving 60 manual analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when a rating agency began sending emails as images, which also improved accuracy; run as a scheduled job on **AWS SageMaker** via EventBridge and Lambda.
- Took the applications from prototype to production by releasing to real users and iterating on what broke; guardrailed responses through **system prompts**, versioned every prompt change in **Git** and had the **Compliance** department review each application before release. Allocated models by cost: open-source **Llama-3** self-hosted on a **GPU EC2** instance for the largest department (Client Service) and stronger **Azure OpenAI** outputs for decision makers, with the ability to switch models.

Python Backend, Microservices & Secure Deployment

- Built the backend of internal **GenAI** applications as **Python FastAPI** microservices (uvicorn) in **Docker**, with a **ReactJS** frontend served by **Nginx** on a separate instance; designed the infrastructure on **AWS** using core services such as **EC2, S3, ECR** and **Lambda**, with separate frontend and backend repositories and containers and a one-click **Jenkins** and **GitLab CI/CD** pipeline for dev and prod.
- Implemented **RBAC** for the GenAI apps with a Python authentication function against the company **LDAP** server; restricted access to company private IPs using **AWS Security Groups**; deployed an **Nginx** reverse proxy,

one of the fixes that took the applications from repeatedly failing to passing the **VAPT** (vulnerability assessment and penetration testing); coordinated with the security team on port whitelisting and development safeguards.

Data Engineering & Client Delivery

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly (fixed income) and monthly (equity); integrated it with legacy data from analysts' Excel files; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.
- Developed **Power BI** dashboards for fortnightly credit report analysis, migrating financial analysts from Excel to Power BI and improving analyst efficiency by **10 hours a week**; then recreated the dashboards in the **ReactJS** frontend so the wider organisation could use the data without Power BI licensing costs. Used **Python Pandas** across data jobs, backend APIs and RAG pipelines.
- Worked directly with stakeholders in **Client Service, Compliance and Investment** to turn requirements into working tools that replace manual work; acted as **de facto team lead for up to 4 interns** in my final year; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; I designed the RAG-based AI assistant and worked on the scoring algorithm and data preprocessing for a tool, built as FastAPI microservices with a React and TypeScript frontend, that ranks cancer cell lines by target inclusion and exclusion genes and returns the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). Behaviour steered by prompts; LLM output treated as untrusted: validated with Pydantic, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, serialised calls, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Architecture study comparing CNN/ViT/ResNet encoders and LSTM/attention decoders for generating image descriptions, benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator. (A self-driving car trained using a **Convolution Neural Network (CNN)** model.)

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2, S3, ECR, Lambda, SageMaker), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite